

FEDERAL URDU UNIVERSITY OF ARTS, SCIENCE AND TECHNOLOGY

Department of Computer Science

PERSONALIZED DRUG REACTION PREDICTOR

(Course Project : Artificial Intelligence)

Submitted By:

Wania Saleem

Seat No: 23122129

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Supervised By:

Ma'am Uzma

Personalized Drug Reaction Predictor

(Rule-Based AI Agent)

1. Problem Identification in Terms of an Agent

Every year, many patients suffer avoidable adverse drug reactions (ADRs) because a prescribed medicine does not suit their personal health profile. For example; an existing allergy, a conflicting medication, age-related sensitivity, or an underlying condition. Doctors and pharmacists cannot always manually cross-check every rule for every patient in real time, so there is a need for an automated assistant that reasons about a specific patient's data and warns of potential drug reactions before the medicine is administered.

This problem is naturally framed as an intelligent agent problem because it involves a system that must:

- **perceive** information about the patient and the proposed drug (its environment),
- **reason** over that information using a defined set of medical rules, and
- **act** by producing a recommendation or warning that changes what the doctor/patient does next.

The agent proposed here **the Personalized Drug Reaction Predictor** is a rule-based agent: it uses an explicit knowledge base of condition-action (IF-THEN) rules built from known drug interactions, allergies, and contra-indications, rather than learning from data. Given a patient's profile and a candidate drug, it senses the relevant facts, matches them against its rule base, and outputs a personalized prediction of the likely reaction and its risk level.

2. Problem Formulation

Input: Patient age, weight, allergies, current medication, and selected drug.

Dataset: Predefined drug-reaction rules containing drug, allergy, medication interaction, and age-related conditions.

i.e,

Patient Age	Allergy	Current Medication	Drug	Rule/Condition	Reaction
25	None	None	Drug A	No allergy or interaction	Safe
30	Penicillin	None	Drug A	Penicillin allergy	Danger
45	None	Blood Thinner	Drug B	Interaction with blood thinner	Danger
60	None	None	Drug C	Age above 55	Caution
35	None	Antibiotic	Drug D	No known interaction	Safe

Algorithm: Rule-Based AI using **IF-THEN matching** to compare input with the dataset.

Output: Safe / Caution / Danger with a short reason.



3. Evaluation in the Context of PEAS

PEAS stands for Performance measure, Environment, Actuators, and Sensors. A standard way to specify a task environment for an agent.

PEAS Element	Description for the Drug Reaction Predictor
Performance Measure	Prediction accuracy, patient safety, speed of response, adherence to medical rule-base/guidelines.
Environment	The patient's personal health profile ,the drug(s) being considered, drug interaction/contra-indication knowledge base, and the user/doctor interface through which data is entered and results are shown.
Actuators	Output/display module that shows the predicted reaction, a risk level (e.g., Safe / Caution / Danger),and reason.
Sensors	Input/intake module that senses (reads) patient data entered by the user or fetched from records: symptoms, current medications, allergies, age, weight, gender, medical history, and the new drug being queried.

4. Environment Properties

The task environment of the Personalized Drug Reaction Predictor can be characterized along the standard dimensions used to analyze intelligent agents, as summarized below.

Property	Classification	Justification
Observability	Partially Observable	The agent only "senses" the data explicitly entered about the patient (allergies, current medicines, age, etc.). It cannot directly observe internal biochemical/genetic factors that are not entered, so its knowledge of the true patient state is incomplete.
Determinism	Deterministic (within the rule base)	Because it is a rule-based system, the same input data will always fire the same rules and always produce the same predicted reaction/output there is no randomness in how the algorithm decides.
Episodic vs. Sequential	Episodic	Each drug-reaction query is treated as a separate, self-contained episode. The prediction for one drug query does not depend on the agent's previous predictions for other queries (though the patient's stored profile persists as data, not as a chain of decisions).
Static vs. Dynamic	Semi-static (mostly static)	The environment (patient profile, drug database) does not change while the agent is computing a single prediction. It changes only between sessions, e.g., when new symptoms appear or a new drug is added, so it is treated as static during each reasoning cycle.
Discrete vs. Continuous	Discrete	Patient attributes, drug names, and risk levels are represented as a finite set of distinct values/categories (e.g., Low/Medium/High risk, Yes/No allergy), and the rule base fires on discrete conditions rather than continuous signals.
Known vs. Unknown	Known	The rules governing drug interactions and reactions are pre-defined by medical knowledge and encoded by the designer, so the "physics" of the environment (which combinations cause which reactions) are known in advance to the agent.
Single vs. Multi-agent	Single-agent	The system acts as one personalized decision-support agent working for a single patient/user at a time; it does not compete or cooperate with other independent agents in the environment.